

The Connect Between Cloud Computing and Android Apps

Android, a Linux-based operating system whose code was released by Google under the Apache licence, is used primarily in touchscreen devices. It has a large developer community, which writes apps that expand the functionality of these devices. It is by far one of the most popular platforms for mobile developers. Android applications developers can now benefit from the power of cloud computing to deliver exciting applications to enterprises.



When studying Android architecture, you can see that the Linux kernel layer provides memory management, security settings, power management, hardware abstraction, etc. The native layer libraries are written in C or C++ and are specific to a particular hardware. Surface manager, the media framework, SQLite, OpenGL, etc., are some of the important native libraries available for Android.

The application framework layer contains the application blocks that directly interact with and manage the device's basic functions such as resource management, voice call management, etc. Content providers, telephony manager, location manager, activity manager and resource manager are some of the important blocks.

The applications layer is the topmost layer in the Android architecture. The SMS client app, the dialler, the Web browser, and contact manager are some of the standard examples. Each Android application can be further subdivided into functional units.

Why cloud computing for Android devices?

Cloud computing has taken the IT world by storm. There are various layers to the Android programming model

that easily fit in with the creation of secure applications specially made for the cloud environment. The open source Android operating system allows complex cloud computing applications to run wherever the user is.

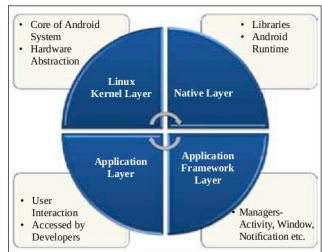


Figure 1: Android architecture